

Scalability & Future Plan

Date	10 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only English-language speech.	Limits multilingual usage.	High
2	Uses local file storage instead of a database.	No long-term record management.	Medium
3	Supports uploaded audio only (no live microphone input).	Limits real-time assessment capability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Streamlit application	Deploy on cloud infrastructure with concurrent user support.
2	Data Storage	Local folders (uploads, reports, temp)	Integrate cloud storage or relational database (PostgreSQL/MySQL).
3	Performance	Local AI model inference	GPU-enabled deployment and optimized model serving.
4	Security	Local application with basic validation	Add authentication, encrypted storage, and secure APIs.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Multi-language speech transcription and evaluation.	3–6 Months	Broader accessibility and support for diverse users.
Phase 3	Real-time microphone recording and live evaluation.	6–9 Months	Improved user experience and interactive assessments
Phase 4	Teacher dashboard, student history, cloud database, and analytics.	9–12 Months	Comprehensive educational assessment platform.